

Solving Absolute Value Linear Equations

Algebra II | Unit 2 | Day 5 | TEK A2.6.E

Objective: I can solve absolute value linear equations by isolating the absolute value and splitting into two cases.

Vocabulary

Isolate — get the absolute value expression _____ on one side before splitting.**Split into two cases** — $|A| = k$ ($k > 0$) becomes $A = k$ _____ $A =$ _____.**Check** — plug each solution back into the _____ equation before accepting it.

Part 1 — The Method

*Step 1: Isolate the absolute value. Step 2: Split into two cases. Step 3: Solve each. Step 4: Check.***Example 1.** $|2x - 3| - 4 = 3$ Isolate: $|2x - 3| =$ _____Split: $2x - 3 = 7$ OR $2x - 3 =$ _____Solve: $x =$ _____ OR $x =$ _____ Solution set: { _____ }

Part 2 — More Examples

Example 2. $|x| + 2 = 5$ Isolate: $|x| =$ _____ Split/Solve: $x =$ _____ or $x =$ _____ (both valid)**You Try.** $|3x + 6| = 12$ $3x + 6 = 12$ or $3x + 6 =$ _____ $x =$ _____ or $x =$ _____

Part 3 — Isolate First, Then Split

Example 3. $2|x + 1| - 3 = 7$ Isolate: $2|x + 1| =$ _____ $\rightarrow |x + 1| =$ _____Split/Solve: $x + 1 = 5$ or $x + 1 =$ _____ $x =$ _____ or $x =$ _____**You Try.** $3|x - 4| + 1 = 13$ $|x - 4| =$ _____ $x =$ _____ or $x =$ _____

Practice

1. $|x - 4| = 9$ $x =$ _____ or $x =$ _____

2. $|5x| - 2 = 18$ $x =$ _____ or $x =$ _____

3. $4|x + 2| = 20$ $x =$ _____ or $x =$ _____

Exit Ticket

Solve $2|x - 5| - 1 = 9$, showing the isolate and split steps.

$|x - 5| =$ _____ $x =$ _____ or $x =$ _____